D12 OSF Pre-Registration (Japanese master)

D12 OSF Pre-Registration — HEXACO 7-Typology Workplace Harassment Microsimulation (Phase 1 + Phase 2)

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0. 本ドキュメントの位置づけ（meta）

0.1 目的

本ドキュメントは、研究計画 v6/v7（Part 8.3 D12）で確定した「Stage 0 コード実装に着手する**前**に OSF 登録する」方針に従い、研究計画から **pre-registration 該当事項のみ抽出して固定する** master draft である。

OSF 提出時には： 1. 本ドキュメントを英訳した英語版を作成 2. OSF Standard Pre-Registration の Web フォームに転記 3. 本ドキュメント（日本語版）を OSF supplementary として PDF で添付 4. 登録日時の DOI を取得し、本ドキュメント Header に記録

0.2 Nosek et al. 2018 PNAS 9 Challenges への適合性確認

Challenge	本研究の状況	対応
C1 手順変更	Simulation のため data collection はなし	N/A
C2 分析時の仮定違反発見	14-cell main は frequentist bootstrap + BCa（仮定 light）。28-cell EB は Beta-Binomial conjugate（method of moments + sensitivity sweep）で robust 化	Section 6.5 で deviation policy 明記
C3 Data Are Preexisting	N=354（harassment）/ N=13,668（clustering）は **既存 IRB 通過済データ**。ただし (a) 7 類型 × gender 14-cell harassment cross-tabulation、(b) 国レベル aggregate prediction、(c) Counterfactual A/B/C simulation outputs は **すべて未観測**	Section 3.1 で「誰が何を observed したか」を明示。partial blinding 状態を honest acknowledge
C4 縦断・大規模多変量データ	N=13,668 clustering data は HEXACO 6 領域 × 13,668 個人の大規模多変量データ。Cluster centroid 抽出は IEEE 掲載済だが、N=354 上の 7 類型 × gender × harassment cross-tabulation は未生成	Section 2.3 で Stage 0–8 を upfront に完全列挙し、cell-level cross-tabulation 生成前にすべての分析 path を固定。Sensitivity sweep（Section 6.4）も網羅的に pre-spec し garden-of-forking-paths を排除
C5 多数実験 / 同一 study 内多検定	単一 design で 7 仮説（H1–H7）を population-level でテスト	Section 6.3 で H2（4 ordinal pairwise）と H4–H7（3 counterfactual tests）に Bonferroni–Holm 補正（family-wise α=.05）。H1 は単一 primary test で補正不要。H7 は composite single test
C6 Program-level null result reporting	本研究は MAPE > 60% でも publication する commitment（D-NEW8）	Section 7 で 5 ケース（success / partial / failure / H7 reversal / B3 < B2 overfitting）に分けて明示 commit
C7 Discovery research（事前期待薄）	Stage 5 (CMV diagnostic)、Stage 4 (B0–B4 baseline hierarchy) は Phase 2 main hypotheses より prior expectation が薄い	これらも **specific decision rule を pre-register**（Section 5.5 decision rule 表、Section 5.6 Harman first-factor < 50% 閾値）。どちらに転んでも pre-spec rule 内で解釈、post-hoc reframing 禁止
C8 Competing predictions	H7 (ΔP_B > ΔP_A AND ΔP_B > ΔP_C) は 3 種介入 strategy を比較する strong-inference test	Nosek 2018 が言うように competing predictions は pre-reg と相性良く strong inference の利点に転化。Section 6.1 で confirmation / partial / reversal 基準を明記
C9 Narrative inference selectivity	Stage 3 sensitivity sweep（V, f1, f2, EB strength, threshold, K）と Stage 8 transportability sweep すべてを pre-reg で固定	Section 6.4 で固定、Section 6.5 deviation policy で post-hoc selectivity 制限、Section 7 で全 pre-registered outcome を方向問わず publish commit

0.3 本 pre-reg と研究計画 v6/v7 の関係

- **研究計画 v6/v7**: 動機・思想・文献根拠を含む **internal document**（L1+L2+L3+L4 すべて）
- **本 pre-reg**: 実装と検証の **predictive commitment** に純化（**L1 描述・予測 only**）
- L3/L4 規範主張（自己責任批判等）は本 pre-reg には含めない（4 層分離原則、研究計画 Part 0.4）

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1. Study Information

1.1 Title

HEXACO 7-Typology Workplace Harassment Microsimulation: Latent Prevalence Prediction and Target Trial Emulation of Personality-Based and Structural Counterfactuals in Japan

短縮形: “HEXACO Harassment Microsim (Phase 1 + Phase 2)”

1.2 Authors

- **Sole author**: Eisuke Tokiwa
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- Co-authors: なし（sole-authored study）

- **Independent methodologist consultation**: pre-reg lock 後・Stage 1 validation 着手前に外部方法論者（**mode B:**

anonymous external methodologist、専門分野は数理生物学 / mathematical biology）に Section 5 (Analysis Plan) のみ review 依頼予定。本人の意向により本 pre-reg および将来の論文 Acknowledgments では **anonymous** として記載（“We thank an anonymous external methodologist with mathematical biology background for review of the analysis plan”）。氏名は author の private record として保持。Munafò et al. 2017 Box 1 CHDI Foundation 例の軽量化適用。

1.3 Description（abstract レベル）

職場ハラスメントは多要因現象であり、組織 stressor（Bowling & Beehr 2006、ρ=.30–.53）、subjective social status（Tsuno et al. 2015、OR=4.21）、業種、法律・社会規範等の効果は personality より大きい。本研究はこれらを認めた上で、**personality contribution を isolate** し、HEXACO 7 類型ベースの確率モデルが日本の職場ハラスメント prevalence をどの程度予測できるかを検証する。

Phase 1（実証 simulation）: N=354 (harassment behavior) と N=13,668 (HEXACO clustering) の既存データから 14-cell（7 類型 × 2 gender）conditional harassment propensity を推定し、bootstrap で日本労働人口（約 6,800 万）にスケールアップした **latent prevalence** を、MHLW 全国統計（**expressed prevalence**）と triangulate する。

Phase 2（介入 counterfactual）: Hernán & Robins (2020) target trial emulation framework 下で、(A) universal HH intervention（Kruse 2014 anchor）、(B) targeted high-risk type intervention（Hudson 2023 anchor、**主軸**）、(C) structural-only intervention（Pruckner 2013 anchor）の 3 種介入を simulate し、population-level harassment 削減量を予測。

LLM は使用しない。すべての mechanism は確率テーブルベースで透明。

1.4 Hypotheses

H1（Phase 1 main）

14-cell（7 type × 2 gender）conditional harassment propensity を population aggregate に scale up した予測値は、MHLW H28（FY2016、pre-law、過去 3 年間 パワハラ** 被害経験 32.5%）に対して MAPE ≤ 30% の精度で再現される。**

- 主 validation target: **MHLW H28 (FY2016)** 「職場のパワーハラスメントに関する実態調査」（過去 3 年間 power harassment prevalence 32.5%、pre-law）
- 副 validation target: **MHLW R2 (FY2020)** 「職場のハラスメントに関する実態調査」（31.4%、transition）、**MHLW R5 (FY2023, published March 2024)** 「令和5年度 職場のハラスメントに関する実態調査」（19.3%、post-law）
- ★ 重要 scope 注記: 本 study が validate するのは **パワハラ (power harassment) の過去 3 年 prevalence** であり、セクハラ・カスハラ・妊娠出産育児ハラスメント等は別 category。MHLW R5 (FY2023) では同サンプル N=8,000 で セクハラ 6.3%、カスハラ 10.8% も測定されているが、本 study の validation target は **パワハラに限定**。
- 国際 baseline: ILO (2022) Asia-Pacific lifetime 19.2%
- 周辺分布検証 (1): **Pasona Research (2022)** N=28,135 全国就業者 20-69 歳、5 年以内 ハラスメント被害 19.7%（生涯 34.6%）— harassment-specific large-N domestic survey として triangulation
- 周辺分布検証 (2): Tsuno et al. (2015) N=1,546 random sample, 30-day prevalence 6.1%

H2（Phase 1 baseline hierarchy）

B0 (random) < B1 (gender only) < B2 (HEXACO 6 領域線形) ≤ B3 (7 typology) ≤ B4 (B3 + age + 業種推定 + 雇用形態) の単調増加が観測される。

- 関心評価: B3 vs B2 の MAPE 差（typology の incrementality）、B4 vs B3 の MAPE 差（personality slice の incrementality）

H3（Phase 1 latent vs expressed gap）

MHLW H28 (FY2016, pre-law、過去 3 年パワハラ 32.5%) と本 simulation の latent prediction の gap は、MHLW R5 (FY2023, post-law、過去 3 年パワハラ 19.3%) との gap より小さい。 すなわち、pre-law condition は latent-proximal、post-law は environmental gating 強。

カスハラ category emergence (R4 改正法 2022 年 4 月施行による中小企業含む全事業主対応義務化、MHLW R5 で初めて 10.8% として独立計測) は、上述の “パワハラ 32.5% → 19.3% 減少” が単なる prevalence 減少ではなく、(a) 法律下で「ハラスメント」と分類される行為の境界自体が変動した（カスハラが新カテゴリとして expressed 化）、(b) 同時に existing パワハラ category の expressed 報告が抑制された、という **複合的な environmental gating の証拠**。Discussion で本研究の latent vs expressed gap が単純な「環境効果」ではなく「category boundary shift × gating strength」の合成であることを honest acknowledge する。

H4（Phase 2 Counterfactual A: Universal HH intervention）

全人口の HH を +0.3 SD shift（Kruse 2014 d=0.71 の保守的 discount）した条件下で、predicted national prevalence が baseline から ΔP_A だけ削減される。

- δ main = +0.3 SD; sensitivity range [0.1, 0.5] SD
- ΔP_A の方向性予測: 削減（負の方向）

H5（Phase 2 Counterfactual B: Targeted intervention、★ 本研究の主軸）

高リスク類型（Cluster 0 = Self-Oriented Independent profile、副候補 Cluster 4, 6）に対する HH +0.4 SD shift（Hudson 2023 self-selected effect の保守的 discount）は、ΔP_B 削減を予測し、ΔP_B / 介入対象人数 比 (cost-effectiveness) は Counterfactual A のそれを上回る。

- δ main = +0.4 SD; sensitivity range [0.2, 0.6] SD
- 主 target: Cluster 0; 副 target: Cluster 4, 6（実装時に IEEE Cluster 番号と CSV 番号を centroid 数値で照合確定）

H6（Phase 2 Counterfactual C: Structural-only intervention）

個人 personality を変えず確率テーブルを 20% 下方修正した条件下で、predicted national prevalence は baseline から削減されるが、その削減量 ΔP_C は Counterfactual B の ΔP_B より小さい。

- effect_C main = 20%; sensitivity range [10%, 30%]（30% は上限、Bezrukova 2016 / Roehling 2018 / Dobbin & Kalev 2018 / Pruckner 2013 の 4 系統 triangulation）
- domain transfer caveat（Pruckner = newspaper honor、本研究 = workplace harassment）を Discussion で明記

H7（Phase 2 main contrast、★ 主たる predictive commitment）

ΔP_B > ΔP_A and ΔP_B > ΔP_C（targeted > universal AND targeted > structural-only at population level）。

- 失敗時（ΔP_B ≤ ΔP_A or ΔP_B ≤ ΔP_C）も publish（Section 7.3 negative result commitment）

Hypothesis 階層整理

H#	Phase	種別	検証指標
H1	1	Confirmatory predictive	MAPE vs MHLW H28 (FY2016) past-3-year power harassment 32.5%
H2	1	Confirmatory ordinal	B0–B4 MAPE 単調
H2.industry	**1**	**Secondary confirmatory**	**B4 industry-stratified subgroup MAPE ≤ 50% against MHLW R5 業種別 (16-26.8% range)** ★ 追加
H3	1	Exploratory descriptive	MHLW H28 gap < MHLW R5 gap
H4	2	Conditional projection	ΔP_A direction & magnitude
H5	2	Conditional projection	ΔP_B direction & cost-eff
H6	2	Conditional projection	ΔP_C direction & magnitude
H7	2	Confirmatory ordinal	ΔP_B > ΔP_A and ΔP_B > ΔP_C

1.5 本研究が 検証しない こと（scope の honest acknowledgment）

- 個人レベル予測（cell N=10–30、pairwise MDE d ≥ 0.92 では individual prediction 不可、D13 power analysis）
- Personality と SSS の独立寄与（N=354 に SSS 直接測定なし、研究計画 Part 1.5）
- 因果方向（cross-sectional design、reverse causation を完全否定不能、Roberts & DelVecchio 2000 の plateau r=.74 が mitigating evidence）
- 介入の長期効果（Roberts 2017 anchor は 24 週時点のみ）
- 西欧 anchor の日本 transportability（Sapouna 2010 cultural moderator 例、Section 4.4 で sensitivity sweep）

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2. Design Plan

2.1 Study Type

Secondary analysis of preexisting data + microsimulation + counterfactual projection

- 新規データ収集なし
- LLM / generative agent 不使用
- Mechanism: 確率テーブル + Monte Carlo bootstrap + Empirical Bayes shrinkage（Beta-Binomial conjugate、method of moments）
- Causal framing: target trial emulation (Hernán & Robins 2020) + structural causal model (Pearl 2009 do-operator)

2.2 Blinding

OSF Standard Pre-Registration の “Blinding” 概念を本研究に適用すると：

項目	状態
個人の HEXACO score	**既に observed**（N=354 / N=13,668、既存 IRB 通過済）
個人の harassment self-report (N=354)	**既に observed**（Tokiwa harassment preprint で集計済）
7 類型 × gender 14-cell harassment cross-tabulation	**未観測**（本 pre-reg で固定する分析仕様で初めて生成）
Stage 1 国レベル aggregate prediction	**未観測**
Counterfactual A/B/C simulation outputs	**未観測**
MHLW survey との MAPE	**未観測**

→ Nosek 2018 PNAS Challenge 3 の strict adherence 下で **“pure” pre-registration** が成立する範囲を明確化。

追加の blinding-equivalent commitment: - Stage 0 コード実装着手以前に本 pre-reg を OSF 登録 - 14-cell cross-tabulation 集計を行う前に inference criteria（Section 6.6）と sensitivity sweep（Section 6.4）を固定 - MHLW survey との比較を行う前に MAPE success/failure threshold（30% / 60%）を固定

2.3 Study Design

2.3.1 Phase 1 Pipeline

```
Stage 0: Type assignment & probability table construction
  ├─ Input: harassment/raw.csv (N=354), clustering/csv/clstr_kmeans_7c.csv (centroids)
  ├─ Step 1: 各 N=354 個人を 7 centroid に Euclidean 最近傍分類 → 7 類型 membership
  ├─ Step 2: 14-cell (7 type × 2 gender) crosstab で harassment binary outcome 集計
  │            (binarization: mean + 0.5 SD per outcome、main; sensitivity: +0.25 / +1.0 SD)
  ├─ Step 3: Bootstrap B=2,000 iter / cell、BCa CI（Efron 1987、DiCiccio & Efron 1996）
  └─ Output: 14-cell propensity table with 95% CI

Stage 1: Population aggregation
  ├─ Input: Stage 0 output + N=13,668 type distribution + MHLW labor force composition
  ├─ Step 1: 13,668 から 7 類型 population proportion を取得
  ├─ Step 2: 厚労省労働力調査の age × gender 分布で reweighting
  ├─ Step 3: Cell-conditional probability × population weights → expected national latent prevalence
  └─ Output: National latent prevalence with bootstrap CI

Stage 2: Validation triangulation
  ├─ Compare national latent power-harassment prediction to MHLW H28 FY2016 (32.5%, primary) / R2 FY2020
(31.4%) / R5 FY2023 (19.3%)
  ├─ Metrics: MAPE (primary), Pearson r, Spearman ρ, KS distance, Wasserstein distance, calibration plot
  └─ Output: Validation report + cell-level prediction error map

Stage 3: Sensitivity sweeps
  ├─ V (victim multiplier) ∈ {2, 3, 4, 5}
  ├─ f1 (turnover rate) ∈ {0.05, 0.10, 0.15, 0.20}
  ├─ f2 (mental disorder rate) ∈ {0.10, 0.20, 0.30}
  ├─ EB shrinkage strength ∈ {0.5×, 1.0×, 2.0×}
  ├─ Binarization threshold ∈ {mean + 0.25 SD, +0.5 SD (main), +1.0 SD}
  ├─ Cluster K ∈ {4, 5, 6, 7 (main, IEEE 掲載済), 8}
  ├─ Role estimation models: (a) personality 線形, (b) tree-based, (c) 文献ベース
  └─ Output: Robustness diagnostic table

Stage 4: Baseline hierarchy comparison (B0–B4)
  ├─ B0: Uniform random
  ├─ B1: Gender only
  ├─ B2: HEXACO 6 領域 linear regression
  ├─ B3: 7 typology + gender (⋆ proposed method)
  ├─ B4: B3 + age + 業種推定 + 雇用形態
  └─ Output: MAPE for each baseline, monotonicity check

Stage 5: CMV diagnostic
  ├─ Harman's single-factor test on N=13,668 personality data (target < 50% variance)
  ├─ Marker variable correction (Lindell & Whitney 2001) using HEXACO Openness as theoretical-marker (low
expected correlation with harassment)
  └─ Output: CMV diagnostic supplementary
```

2.3.2 Phase 2 Pipeline

```
Stage 6: Target trial emulation specification
  ├─ Target trial protocol (PICO + duration) for each counterfactual
  ├─ Eligibility criteria: 日本労働者 20–64 歳
  ├─ Treatment strategies: A (universal HH +δ SD), B (targeted HH +δ SD on Cluster 0/4/6), C (structural -
effect_C × cell prob)
  ├─ Assignment: simulated random
  ├─ Outcome: population-level harassment prevalence
  ├─ Follow-up: 24 weeks (Roberts 2017 anchor)
  └─ Output: 4 identifying assumptions explicit (exchangeability, positivity, consistency,
transportability)

Stage 7: Counterfactual simulation
  ├─ A (Kruse 2014 anchor):  shift全 N の HH by +δ SD, re-classify cluster, re-estimate propensity,
aggregate to national
  ├─ B (Hudson 2023 anchor): shift Cluster 0/4/6 individuals' HH by +δ SD only
  ├─ C (Pruckner 2013 + Bezrukova 2016 + Dobbin & Kalev 2018 + Roehling 2018 triangulation): multiply
cell-conditional probability by (1 − effect_C)
  └─ Output: ΔP_A, ΔP_B, ΔP_C with bootstrap CI

Stage 8: Counterfactual sensitivity
  ├─ A: δ ∈ [0.1, 0.5] SD sweep
  ├─ B: δ ∈ [0.2, 0.6] SD sweep
  ├─ C: effect_C ∈ [0.10, 0.30] sweep
  ├─ Transportability: 西欧 anchor effect × {0.3, 0.5, 0.7, 1.0} (Sapouna 2010 / Nielsen 2017 cultural
moderator)
  └─ Output: Robustness table
```

2.4 Randomization

物理的 randomization なし（observational + simulation）。Bootstrap resample および Monte Carlo は **固定 seed**（NumPy `default_rng(seed=20260429)`）で deterministic に再現。Seed value は本 pre-reg で固定。

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3. Sampling Plan

3.1 Existing Data Statement (Nosek 2018 PNAS Challenge 3 strict adherence)

3.1.1 既に収集済かつ part of authors により observed

- **N=354 harassment data** (`harassment/raw.csv`):
  - Tokiwa harassment preprint で HEXACO + Dark Triad → harassment の HC3-robust regression 集計済
  - **Observed by authors at individual level**: HEXACO 6 領域 score, Dark Triad 3 score, power harassment scale, gender harassment scale, age, gender, area
- **N=13,668 clustering data**:
  - Tokiwa clustering paper（IEEE 掲載済）で 7 類型 centroid と分布が確定
  - **Observed by authors at aggregate level**: 7 centroid (HEXACO 6 領域)、cluster proportion、aggregate descriptives

3.1.2 未観測（本 pre-reg で初めて固定する分析）

- N=354 を 7 centroid に最近傍分類した結果の 7 類型 membership distribution
- 7 type × 2 gender 14-cell crosstab の harassment binary outcome
- Cell-conditional propensity の bootstrap BCa CI
- 28-cell EB-shrunken estimate
- 国レベル aggregate latent prevalence
- MHLW 全国統計との MAPE
- Counterfactual A/B/C の ΔP estimate

3.1.3 Pre-registration “purity” の honest acknowledgment

著者は N=354 の HEXACO ↔ harassment associations を harassment preprint の HC3 regression で部分観測している。本 pre-reg で固定する **type-conditional cell-level propensity table** および **国レベル aggregate prediction** は当該 regression からは直接導出不可だが、**完全 blind ではない**。

→ Nosek 2018 Challenge 3 の “pure pre-registration” は理論的最善であるが、本研究は **partial blinding 状態** であることを明示し、findings の interpretation で「7 類型 cross-tab は preregistered だが、HEXACO 領域レベルの associations は

exploratory replication of prior work」と区別する。

3.2 Sample Size

3.2.1 Cell-level (Phase 1 main analysis)

D13 power analysis (`simulation/docs/power_analysis/D13_power_analysis.md`) より：

項目	値
Cell 数（main）	14 (7 type × 2 gender)
Cell N 最小 / 最大 / 中央値	10 / 70 / 18
Cell N < 10	0 cells (0%)
Cell N < 20	7 cells (50%)
One-sample MDE (Cohen's d) 中央値 / 最大	0.67 / 0.89
Pairwise MDE (Cohen's d) 中央値 / 最大	0.92 / 1.25
Binary rate CI ±half-width 中央値	±13 percentage points (power harassment)
Bootstrap iterations	2,000 / cell
Bootstrap CI method	BCa (Efron 1987)

3.2.2 Cell-level (Phase 1 sensitivity)

項目	値
Cell 数（sensitivity）	28 (7 type × 2 gender × 2 role)
Cell N < 10	16 cells (57%)
Cell N = 0	4 cells
Cell N ≤ 3	9 cells
必須対応	Empirical Bayes shrinkage (Beta-Binomial conjugate, Casella 1985 + Clayton & Kaldor 1987 + Efron 2014 + Greenland 2000)

3.2.3 National-level (Phase 1 + Phase 2)

母集団 6,800 万（厚労省労働力調査 base）。Cell-conditional probability に population weights を適用、bootstrap で aggregate CI を生成。

3.3 Sample Size Rationale

- D13 で **N ≥ 250 (Funder & Ozer 2019 stable r estimation 推奨)** を aggregate analysis で満たす（N=354 ≥ 250）
- 14-cell main で **N ≥ 10 / cell 全充足** → shrinkage なしで bootstrap 推定可能
- Pairwise MDE d ≥ 0.92 は "very large effect" (Cohen 1988) / "rarely found in replication" (Funder & Ozer 2019) → **cell-level pairwise inference は avoid、aggregate-level に focus** という pre-registered limitation

3.4 Stopping Rule

新規データ収集なしのため stopping rule は適用外。**Sensitivity sweep の追加・拡張は本 pre-reg で固定した範囲（Section 6.4）に限定**。Pre-reg 後の追加 sensitivity は exploratory として明示し、confirmatory inference には用いない。

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4. Variables

4.1 Manipulated Variables (counterfactual operators)

変数	定義	適用範囲	Main 値	Sensitivity range	出典 anchor
δ_A（universal HH shift）	全 N の HH score を δ × SD(HH) shift	All N=354 → re-classify cluster	**+0.3 SD**	[0.1, 0.5] SD	Kruse 2014 d=0.71 を保守的 discount
δ_B（targeted HH shift）	Cluster 0/4/6 の N のみ HH score を δ × SD(HH) shift	Cluster 0 主、Cluster 4, 6 副	**+0.4 SD**	[0.2, 0.6] SD	Hudson 2023 self-selected effect の保守的 discount
effect_C（structural reduction）	Cell-conditional probability に (1 − effect_C) を乗じる	All 14 cells	**0.20**	[0.10, 0.30]	Pruckner 2013 + Bezrukova 2016 + Roehling 2018 + Dobbin & Kalev 2018 triangulation
transportability_factor	Phase 2 anchor effect × factor で日本適用	All counterfactuals	1.0 (main)	[0.3, 0.5, 0.7, 1.0]	Sapouna 2010 / Nielsen 2017

4.2 Measured Variables (existing observed data)

4.2.1 個人 level (N=354 harassment data)

変数	種別	操作的定義	出典
HEXACO 6 領域	Continuous (Likert 1–5 mean)	Wakabayashi 2014 Japanese HEXACO-60	`harassment/raw.csv`
Dark Triad 3	Continuous	Shimotsukasa & Oshio 2017 SD3-J	`harassment/raw.csv`
Power harassment	Continuous (item mean) → binarized at mean+0.5 SD	Tou et al. 2017 Workplace Power Harassment Scale	`harassment/raw.csv`
Gender harassment	Continuous → binarized at mean+0.5 SD	Kobayashi & Tanaka 2010	`harassment/raw.csv`
Age	Continuous (years)	Self-report	`harassment/raw.csv`
Gender	Binary (0/1, n=133/220)	Self-report	`harassment/raw.csv`
Area	Categorical	Self-report	`harassment/raw.csv`

4.2.2 個人 level (N=13,668 clustering data)

変数	種別	用途
HEXACO 6 領域	Continuous	Centroid 抽出（既に集計済、`clustering/csv/clstr_kmeans_7c.csv`）
Cluster proportion	Categorical 7-type	Population scaling weight

4.2.3 集団 level (MHLW + 民間 large-N、external validation target)

変数	データソース	役割
過去 3 年間 power harassment 被害経験率	**MHLW H28 (FY2016, pre-law) 32.5% ★ primary**, MHLW R2 (FY2020, transition) 31.4%, **MHLW R5 (FY2023, post-law, published 2024-03) 19.3%**	National validation target
業種別 power harassment past-3-year 経験率	**MHLW R5 (FY2023) 業種別: 建設 26.8%, 複合 22.5%, 教育/医療福祉 20.6%, …, 教育 16.9%（業種範囲 16-26.8%、N=8,000 一般 sample）**	**B4 baseline subgroup validation (H2.industry)**
その他ハラスメント category 比較	MHLW R5 セクハラ 6.3% (女 8.9% / 男 3.9%), カスハラ 10.8%（環境変動 framing 用）	latent vs expressed gap framing (H3)
5 年以内ハラスメント被害経験率（民間）	**Pasona Research (2022) 「職場のハラスメントについての定量調査」N=28,135、5 年以内 19.7%（生涯 34.6%）、業種別 16.9-22.9%**	周辺分布検証 (large-N harassment-specific)
30 日 prevalence (random sample)	Tsuno et al. 2015 N=1,546 (6.1%)	周辺分布検証 (national-rep)
国際 baseline	ILO 2022 Asia-Pacific lifetime 19.2%	比較参照
f1 anchor (ハラスメント被害者の離職率)	**Pasona (2022) ハラスメント被害者 5 年以内離職率 10.3% (業種別 6.3-13.3%) ★ primary**; **MHLW R4 雇用動向調査 (FY2022)** 転職入職者前職離職理由「職場の人間関係」男 8.3% / 女 9.4% (upper bound)	f1 sensitivity sweep [0.05, 0.10, 0.15, 0.20] の empirical anchor
f1 macro cross-check	Pasona (2022) 全国年間ハラスメント離職者推定 86.5 万人（うち暗数 57.3 万人 = 66%），全離職者中比率 86.5/717.3 = 12.1%	Stage 2 chain output sanity check (Section 5.4)
メンタル疾患発症率	厚労省労働安全衛生調査 + Tsuno & Tabuchi 2022 PR=3.20	f2 anchor

4.3 Indices (derived)

Index	定義	算出
7-type membership	Each N=354 individual の最近傍 centroid (Euclidean, HEXACO 6 領域)	Stage 0 で生成
Cell ID (14-cell)	type ∈ {0..6} × gender ∈ {0,1}	Stage 0 で生成
Cell ID (28-cell)	type × gender × role ∈ {0,1}	Sensitivity 用
Role probability	Continuous, predicted from C + 0.5·X composite (top 15% → manager)	Stage 0; D1 sensitivity で 3 モデル比較
National latent prevalence	Σ_cell (cell_propensity × cell_population_weight)	Stage 1
MAPE	mean(	predicted − observed
ΔP_x (counterfactual reduction)	predicted_baseline − predicted_counterfactual_x	Stage 7
Cost-effectiveness ratio	ΔP_x / N_treated_x	Stage 7, Phase 2

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5. Analysis Plan (statistical models)

5.1 Phase 1 Stage 0: Cell-level propensity estimation

For each cell c ∈ {1..14} (7 type × 2 gender): - X_c = number of “harassment perpetrator” (binarized at mean+0.5 SD per outcome) in cell c - N_c = total N in cell c - Observed propensity p̂_c = X_c / N_c - Bootstrap distribution: B = 2,000 BCa resamples / cell (Efron 1987 J Am Stat Assoc; DiCiccio & Efron 1996 Stat Sci) - BCa correction: bias z_0 + acceleration a from jackknife

BCa numerical stability fallback chain (★ added v2, anonymous methodologist consultation): - BCa is the primary CI method. - If acceleration parameter |a| > 10 (jackknife instability, typically caused by binary outcome with rate near 0/N or N/N at small N) OR BCa computation raises numerical error (division by zero, NaN propagation), automatically fallback to **Bias-Corrected (BC) bootstrap** (z_0 correction only, no acceleration). - If BC bootstrap also fails (rare; sample distribution too degenerate), fallback to **percentile bootstrap** (no correction). - All cells record which CI method was

actually used; the supplementary table reports per-cell method (BCa / BC / percentile). - Aggregate-level (Stage 1, Stage 2) bootstrap propagation inherits the cell-level method choice automatically.

Output: 14-cell table with point estimate $\hat{p}_c$ and 95% CI $[\hat{p}_c^{lo}, \hat{p}_c^{hi}]$ (method = BCa / BC / percentile recorded per cell).

5.2 Phase 1 Stage 0 (sensitivity): 28-cell EB shrinkage

Beta-Binomial conjugate (Casella 1985 + Clayton & Kaldor 1987 + Efron 2014 Stat Sci + Greenland 2000 IJE):

1. **Method of moments で hyper-prior 推定** (from 14-cell main):
   - $\hat{\mu} = \text{mean}(\hat{p}_j)$, $\hat{\sigma}^2 = \text{var}(\hat{p}_j)$ for j=1..14
   - $\hat{\alpha} = \hat{\mu} \times [\hat{\mu}(1-\hat{\mu})/\hat{\sigma}^2 - 1]$
   - $\beta = (1-\hat{\mu}) \times [\hat{\mu}(1-\hat{\mu})/\hat{\sigma}^2 - 1]$
2. **28-cell posterior** (each cell k):
   - $E[p_k \mid X_k, N_k] = (\hat{\alpha} + X_k) / (\hat{\alpha} + \beta + N_k)$
   - 95% credible interval from $\text{Beta}(\hat{\alpha} + X_k, \beta + N_k - X_k)$ quantiles
3. **Strength sensitivity sweep** (★ pre-registered):
   - scale ∈ {0.5×, 1.0× (main), 2.0×}: hyper-parameters $(\hat{\alpha}, \beta)$ を scale で乗じて weak / medium / strong shrinkage を比較

MoM rejection decision rule (★ added v2, pre-specified, anonymous methodologist consultation):

The pre-reg fixes the following hard rule for choosing between MoM and Stan hierarchical Bayes as the **primary** 28-cell estimator:

- **REJECT MoM, switch to Stan** if EITHER of:
  - $\hat{\sigma}^2 / [\hat{\mu}(1-\hat{\mu})] < 0.05$ (variance too small relative to maximum-possible binomial variance; MoM produces extreme $\hat{\alpha}, \beta$ with overshrink risk)
  - $\max(\hat{\alpha}, \beta) > 100$ (pseudo-count exceeds 7× the median cell N=14; MoM dominates likelihood signal)
- **ACCEPT MoM as primary** if both conditions clear.
- The threshold values 0.05 and 100 are FIXED by this v2 preregistration; no post-hoc adjustment is permitted.

In ALL cases (regardless of acceptance / rejection), THREE estimators are computed and reported (★ added v2 triangulation requirement):

Estimator	Status	Reporting
MoM (Beta-Binomial conjugate)	Primary IF accepted	Main text + supplementary
Marginal MLE (numerical maximization)	Always run	Supplementary (triangulation)
Stan hierarchical Bayes (HMC, brms)	Primary IF MoM rejected	Main text (if primary) + supplementary always

Cross-check diagnostic (★ added v2, methodologist suggestion): in addition to the $\hat{\sigma}^2 / [\hat{\mu}(1-\hat{\mu})]$ threshold, the supplementary reports the alternative diagnostic **$(\hat{\alpha} + \beta)$ / median cell N** as a second indicator of overshrink concern. Readers can verify that the threshold-based switch does not produce qualitatively different conclusions.

MoM threshold sensitivity caveat (★ added v2, methodologist note): $\hat{\sigma}^2$ is itself a sample estimate from N=14 cells with sampling variability. To address this, the supplementary reports MoM and Stan results in parallel even when MoM is "accepted" by the rule, demonstrating robustness across the threshold boundary.

Diagnostic plots (★ existing, retained): $\hat{\mu} \pm$ SE plot; shrunken vs raw scatter; MoM-vs-Stan agreement scatter (★ added v2).

5.3 Phase 1 Stage 1: Population aggregation

For target validation period t ∈ {2016, 2020, 2024}: - W_c (cell weight) = MHLW labor force population × cluster proportion (from N=13,668) × gender proportion × age weight - National latent prevalence: $\hat{P}_t = \Sigma_c (\hat{p}_c \times W_c) / \Sigma_c W_c$ - Bootstrap CI for $\hat{P}_t$: 2,000 iter, BCa

Demographic reweighting source: 厚労省労働力調査（age × gender × 雇用形態）

5.4 Phase 1 Stage 2: Validation triangulation

Bootstrap MAPE CI computation procedure (★ added v2, anonymous methodologist consultation, pre-specified):

For each bootstrap iteration b ∈ {1..2,000}:

1. Resample N=354 with replacement, **stratified by 14 cell** (preserving cell membership marginals; this is the cell-

level bootstrap, NOT independent resampling of population weights).

2. Re-classify each resampled individual to the nearest of 7 centroids (centroids fixed from N=13,668; not bootstrapped).
3. Re-compute each cell's binary harassment outcome propensity $\hat{p}_c^{(b)}$ on the resampled data, applying the BCa numerical stability fallback chain (Section 5.1) at the cell level.
4. Apply population weights W_c (from MHLW Labor Force Survey, FIXED across iterations — not bootstrapped) to compute the resampled national latent prevalence: $\hat{P}_t^{(b)} = \Sigma_c (\hat{p}_c^{(b)} \times W_c) / \Sigma_c W_c$
5. Compute the resampled MAPE: $MAPE^{(b)} = |\hat{P}_t^{(b)} - MHLW_observed_t| / MHLW_observed_t \times 100$ where $MHLW_observed_t$ is the FIXED point estimate from MHLW survey for period t (no bootstrap of MHLW data).
6. Record $MAPE^{(b)}$.

After all B = 2,000 iterations: - **Point estimate MAPE** is computed on the original (non-bootstrapped) data, NOT the mean of $MAPE^{(b)}$. The bootstrap distribution informs only the CI, not the point. - **95% BCa CI for MAPE** is constructed from the empirical distribution of $\{MAPE^{(b)}\}$ using bias correction z_0 + acceleration a from jackknife (BCa fallback chain applies if numerical instability detected at the MAPE level).

Both point estimate and 95% BCa CI are reported for each validation period $t \in \{FY2016, FY2020, FY2023\}$. Identical procedure applies to all baseline models B0–B4.

4-tier judgment hierarchy (★ added v2, primary inference criterion, pre-registered):

Tier	Condition	Interpretation
Strict SUCCESS	Point MAPE ≤ 30% **AND** 95% bootstrap CI upper bound ≤ 30%	Strong confirmatory result; CI rules out PARTIAL or FAILURE regions
Standard SUCCESS	Point MAPE ≤ 30% (CI permitted to overlap 30%–60% PARTIAL region)	Weak confirmatory; Discussion explicitly notes "Pre-registered ambiguity Tier"
PARTIAL SUCCESS	30% < point MAPE ≤ 60%	Mixed evidence; CI flagged in Discussion
FAILURE	Point MAPE > 60%	Publish as failure-mode discovery (Section 7.3)

Tier assignment for **H1 primary** is based on FY2016 validation (against MHLW H28 32.5%). FY2020 and FY2023 receive independent tier classifications and are reported as secondary.

Tier-specific reporting requirements (★ added v2, pre-registered):

- **Strict SUCCESS**: "Strict SUCCESS achieved" allowed in headline / abstract without further qualification.
- **Standard SUCCESS**: "Standard SUCCESS" must be qualified with explicit CI ambiguity statement, e.g., "Standard SUCCESS achieved (point MAPE = X%, 95% bootstrap CI = [Y%, Z%]). The CI overlapped the PARTIAL region [30%, 60%], so the inferential strength is limited; the result falls in the **Pre-registered ambiguity Tier**, and confirmatory claims are correspondingly weakened."
- **PARTIAL SUCCESS**: standard PARTIAL framing with CI reported.
- **FAILURE**: standard FAILURE framing per Section 7.3.

All Tiers continue to satisfy Section 7 negative-result publication commitment (D-NEW8). Post-hoc tier-threshold revision is PROHIBITED; v3 OSF registration with public diff is required for any change.

Secondary metrics (descriptive): - Pearson r between cell-level $\hat{p}_c$ and MHLW R5 (FY2023) subgroup rates - Spearman ρ (rank correlation) - KS distance (distribution shape) - Wasserstein distance (earth mover) - Calibration plot (cell predicted vs observed)

Subgroup MAPE (★ pre-registered): - Gender × age band → subgroup MAPE で failure mode の可視化

Triangulation against marginal-distribution surveys (★ pre-registered, descriptive): - **Pasona (2022) N=28,135 5-year prevalence 19.7%**: report Pearson r between simulation cell-level prediction and Pasona industry-stratified rates (16.9-22.9%); descriptive only, no SUCCESS / FAILURE threshold - Tsuno et al. (2015) N=1,546 30-day prevalence 6.1%: reference period correction discussed in Methods (cannot be directly compared to past-3-year prediction) - ILO (2022) 19.2% Asia-Pacific lifetime: report difference (Δ between predicted past-3-year and reported lifetime) for context

Stage 2 chain output sanity check (★ pre-registered, soft criterion): - Compute predicted annual harassment-induced turnover via V × f1 chain (Stage 2): Σ_cell (predicted perpetrators × V × f1) - Compare against **Pasona (2022) 全国年間ハラスメント離職者推定 86.5 万人** (with 暗数 57.3 万人 = 66%) - Soft criterion: predicted annual turnover within 50-200% of Pasona estimate (i.e., 43-173 万人) — outside this range flagged for failure-mode discussion - This is a SECONDARY criterion (does not affect H1 SUCCESS / FAILURE)

5.5 Phase 1 Stage 4: Baseline hierarchy

For each baseline $B \in \{B0, B1, B2, B3 (proposed), B4\}$: - Train: 同一 N=354 - Predict national prevalence - Compute MAPE

vs MHLW H28 (FY2016) past-3-year power harassment 32.5% - Pre-registered ordinal hypothesis (H2): MAPE_B0 ≥ MAPE_B1 ≥ MAPE_B2 ≥ MAPE_B3 ≥ MAPE_B4

Models: - **B0**: Uniform p_random = MHLW H28 (FY2016) grand mean - **B1**: gender-only logistic (P̂(harassment | gender)) - **B2**: HEXACO 6 領域 linear (logistic regression, all 6 domains) - **B3** (proposed): 7 type × gender cell-conditional (本研究 main) - **B4**: 7 type + age + **業種 (estimated)** + 雇用形態 cell-conditional

Industry estimation specification for B4 (★ pre-registered, ★ Item 5): - N=354 does **not contain direct industry data**. B4's "業種" is **estimated probabilistically** from age × gender × employment type joint distribution against MHLW 労働力調査 industry × demographic crosstabs. - Estimation method (★ pre-registered): for each individual i with (age_i, gender_i, employment_i), assign a 16-bucket industry probability vector P(industry_j | age_i, gender_i, employment_i) using MHLW labor force data 2022. The cell prediction for B4 is then weighted by this probability vector. - 16 industry buckets follow MHLW 日本標準産業分類 main divisions (matching MHLW R5 industry-stratified harassment data structure). - This estimation is **inherently noisy** (individual industry is not observed); B4's industry-level predictions are therefore **expected to have wide CIs**.

Decision rule (overall MAPE): - B3 > B2 → "typology が線形を上回る情報追加" - B3 ≈ B2 → "typology は線形より優位ではない"（finding として report） - B3 < B2 → "typology が overfit"（critical finding として report） - B4 ≫ B3 → "personality slice 単独不十分、周辺変数必要" - B4 ≈ B3 → "personality typology が周辺変数を informationally subsume"

Industry-stratified validation (H2.industry) (★ pre-registered, ★ Item 5): - B4 industry-stratified predictions (16 buckets) are compared to **MHLW R5 (FY2023) industry-stratified past-3-year power harassment rates** (16-26.8% range) - **Subgroup MAPE threshold (relaxed)**: 50% (more lenient than overall H1 30%, given industry estimation noise; 業種 is estimated, not observed) - Decision rule: - Industry-MAPE ≤ 50% → CONFIRMS that personality + estimated industry can predict industry-level patterns - Industry-MAPE > 50% → REPORT as honest limitation (industry estimation noise dominates personality signal at industry level) - This is a **secondary criterion** to H1; failure does NOT trigger H1 failure.

5.6 Phase 1 Stage 5: CMV diagnostic

Harman's single-factor test (Podsakoff et al. 2003 J Appl Psychol): - N=13,668 personality data 全 HEXACO items を 1 factor unrotated EFA - 第 1 因子 variance explained < 50% → CMV concern 限定的（pre-registered threshold）

Marker variable correction (Lindell & Whitney 2001 J Appl Psychol): - HEXACO Openness を theoretical-marker として selection - Marker と target correlations の partial-out で adjusted associations を report

5.7 Phase 2 Stage 6–8: Counterfactual estimation

5.7.1 Target trial emulation specification

各 counterfactual に対し PICO + duration を pre-register:

Element	Counterfactual A	Counterfactual B (★ main)	Counterfactual C
P (population)	日本労働者 20–64 歳	日本労働者 20–64 歳	日本労働者 20–64 歳
I (intervention)	Universal HH +δ_A SD	Targeted (Cluster 0/4/6) HH +δ_B SD	Structural × (1 − effect_C) on cell prob
C (control)	Pre-intervention baseline	Pre-intervention baseline	Pre-intervention baseline
O (outcome)	National harassment prevalence	National harassment prevalence + cost-effectiveness	National harassment prevalence
Duration	24 weeks (Roberts 2017)	24 weeks	24 weeks
Anchor	Kruse 2014 d=0.71	Hudson 2023 b=.03/week	Pruckner 2013 + Bezrukova 2016 + Roehling 2018 + Dobbin & Kalev 2018

5.7.2 Pearl 2009 do-operator notation

- Counterfactual A: do(HH := HH + δ_A × SD(HH)) for all individuals
- Counterfactual B: do(HH := HH + δ_B × SD(HH)) for individuals in Cluster ∈ {0, 4, 6}
- Counterfactual C: do(p_c := p_c × (1 − effect_C)) for all cells c

5.7.3 Estimation

For each counterfactual x ∈ {A, B, C}: - Apply do-operator to N=354 / cell-prob table per spec - Re-run Stage 0 → Stage 1 (Stage 2 validation 不要、prediction のみ) - ΔP_x = P̂_baseline − P̂_x - Bootstrap CI for ΔP_x (2,000 iter, propagate cell-level uncertainty) - Cost-effectiveness for B: ΔP_B / |Cluster 0 ∪ 4 ∪ 6 in population|

5.7.4 Identifying assumptions (Hernán & Robins 2020)

Discussion で **明示的に honest 評価**:

1. **Exchangeability**: Y^a ⊥⊥ A | L
   - Violation risk: 文化・組織風土の unmeasured confounding
   - Mitigation: B4 baseline で周辺変数調整、sensitivity sweep
2. **Positivity**: P(A=a | L=l) > 0 for all l
   - Violation risk: Cluster 6 dominant 32% で intervention coverage 不均一
   - Mitigation: Cluster 6 を意図的に counterfactual analysis から除外しない
3. **Consistency**: observed Y when A=a equals Y^a (no interference between agents)
   - Violation risk: 職場での peer effect、displacement
   - Mitigation: Counterfactual C の displacement を Discussion で明示
4. **Transportability**: anchor study population effect が target Japanese workforce に transport
   - Violation risk: Kruse / Hudson / Pruckner はすべて西欧 / 米国 sample
   - Mitigation: Section 5.8 transportability sensitivity sweep + Sapouna 2010 / Nielsen 2017 cultural moderator 引用

5.8 Phase 2 Stage 8: Transportability sensitivity

Factor	Range	解釈
0.3×	Strong cultural attenuation (Sapouna 2010 UK→Germany null worst case)	保守的 lower bound
0.5×	Moderate attenuation (Nielsen 2017 Asia/Oceania Neuroticism r=.16 vs Europe .33 比)	“expected” attenuation
0.7×	Mild attenuation	Optimistic
1.0× (main)	No attenuation (anchor effect = Japan effect)	Reference

→ Conclusions の robustness を transportability_factor に対して report

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6. Inference Criteria & Sensitivity Master Table

6.1 Pre-registered inference criteria summary

H#	Criterion	Threshold	Decision
H1 (Strict SUCCESS) ★ v2	Point MAPE(P̂_FY2016, MHLW H28 FY2016 32.5%) **AND** 95% BCa CI upper bound	both ≤ 30%	Strong confirmatory; "Strict SUCCESS achieved" allowed without qualification
H1 (Standard SUCCESS) ★ v2	Point MAPE	≤ 30% (CI may overlap PARTIAL region)	Weak confirmatory; **Pre-registered ambiguity Tier** note required (Section 5.4)
H1 (Partial)	Point MAPE	30 < x ≤ 60%	Mixed evidence, CI reported in Discussion
H1 (Failure)	Point MAPE	> 60%	Publish as failure-mode discovery (Section 7.3)
H2	MAPE_B0 ≥ MAPE_B1 ≥ MAPE_B2 ≥ MAPE_B3 ≥ MAPE_B4	Strict monotonicity	Pre-registered direction confirmed if ≥ 3 pairwise inequalities hold
H2.industry ★	B4 industry-stratified (16-bucket) MAPE vs MHLW R5 (FY2023) industry-stratified power harassment 16-26.8%	≤ 50% (relaxed; 業種 estimated)	CONFIRMED if ≤ 50%; honest LIMITATION REPORT if > 50%. Secondary criterion (does not affect H1 SUCCESS / FAILURE).
H3	gap(H28 FY2016) < gap(R5 FY2023)	Direction	Confirmed if MAPE_FY2016 < MAPE_FY2023
H4	sign(ΔP_A)	< 0 (削減)	Confirmed if 95% CI excludes 0 in negative direction
H5	sign(ΔP_B) and ΔP_B / N_treated > ΔP_A / N_total	Cost-eff	Confirmed if both conditions hold
H6	sign(ΔP_C)	< 0	Confirmed if 95% CI excludes 0
H7	ΔP_B > ΔP_A AND ΔP_B > ΔP_C	Magnitude ranking	Confirmed if both inequalities at point estimate; uncertain if 95% CI overlap

6.2 Failure mode commitment

- **H1 failure** (MAPE > 60%): Publish as failure mode discovery (Doc 1 戦略 2 + Nosek 2018 Challenge 6)
- **H2 reversal** (B3 < B2): Publish as critical finding (typology overfitting)
- **H7 reversal** (ΔP_B ≤ ΔP_A or ΔP_B ≤ ΔP_C): Publish; main thesis claim 修正

6.3 Multiple comparison correction

- Phase 1 H1: Single primary test, no correction needed
- Phase 1 H2: 4 ordinal pairwise comparisons (B0-B1, B1-B2, B2-B3, B3-B4) → Bonferroni-Holm at α = .05 family-wise
- Phase 2 H4–H7: 3 counterfactuals × 1 main test each = 3 tests → Bonferroni-Holm at α = .05
- H7 ranking: Already a single composite test (no further correction)

6.4 Pre-registered sensitivity sweep master table (★ Stage 3 / Stage 8)

Phase 1 sensitivity (all combinations report MAPE table):

Parameter	Main	Sweep range	Stage
V (victim multiplier)	3	{2, 3, 4, 5}	3
f1 (harassment-induced turnover rate)	0.10 (close to Pasona 2022 empirical 10.3%)	{0.05, 0.10, 0.15, 0.20} (covers Pasona industry range 6.3-13.3% + MHLW R4 「人間関係」離職 8.3-9.4%)	3
f2 (mental disorder rate)	0.20	{0.10, 0.20, 0.30}	3
EB shrinkage scale	1.0×	{0.5×, 1.0×, 2.0×}	0 (28-cell)
Binarization threshold	mean+0.5 SD	{mean+0.25 SD, +0.5 SD, +1.0 SD}	0
Cluster K	7 (IEEE 掲載済 main)	{4, 5, 6, 7, 8}	0
Role estimation	C+0.5X composite top 15%	{(a) linear, (b) tree-based, (c) literature}	0 (28-cell)
Bootstrap iterations	2,000	(fixed)	All

Phase 2 sensitivity:

Parameter	Main	Sweep range	Stage
δ_A (universal HH shift)	+0.3 SD	[0.1, 0.5] SD step 0.1	8
δ_B (targeted HH shift)	+0.4 SD	[0.2, 0.6] SD step 0.1	8
effect_C (structural reduction)	0.20	[0.10, 0.30] step 0.05	8
transportability_factor	1.0×	{0.3×, 0.5×, 0.7×, 1.0×}	8
Target type set (B)	{Cluster 0, 4, 6}	{Cluster 0 only}, {0+4}, {0+4+6} (main), {0+4+6+others}	8

Reporting: 全 sensitivity 結果は supplementary table として open data に publish (Section 8 reproducibility)。Main text は main 値のみを quantitative claim で使用、sensitivity は robustness 確認の qualitative statement に限定。

6.5 Deviation policy (Nosek 2018 Challenge 1, 2 対応)

Pre-reg からの逸脱は以下に分類して report:

1. **Level 0 (no deviation)**: 仕様通り実装
2. **Level 1 (minor specification clarification)**: 例 — bootstrap seed の lookup table での明示化、CI 計算 library version の固定化等。Methods で 1 行記録
3. **Level 2 (data-driven adjustment)**: 例 — Method of Moments で $\hat{\alpha}, \beta$ が発散 → MLE / hierarchical Bayes に切替。Discussion 1 段落で justification
4. **Level 3 (analysis plan revision)**: 例 — 14-cell pairwise inference を行う等の重大変更 → **本 pre-reg を update し、版 v2 を OSF に新規登録**。Original v1 と diff を public 化

→ Level 2 以上の逸脱を Discussion の専用 subsection “Deviations from Pre-Registration” で comprehensive 報告。

6.6 Inference data lock-in commitments

- **MHLW survey との比較** は本 pre-reg を OSF 登録した後に初めて実施
- **MAPE 計算** は本 pre-reg で固定された binarization threshold (mean+0.5 SD) と population weights で先行実施
- **MAPE threshold (30% / 60%)** の post-hoc 変更は禁止

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7. Negative Result & Failure Mode Publication Commitment (D-NEW8)

Anchor: **Nosek et al. 2018 PNAS Challenge 6** (program-level null result reporting) + Doc 1 戦略 2 (失敗の主題化)

7.1 公表 commitment 内容

以下のすべてのケースで投稿論文として publish する:

1. **H1 SUCCESS** (MAPE ≤ 30%): 通常論文として投稿
2. **H1 PARTIAL SUCCESS** (30 < MAPE ≤ 60%): “partial validation” 論文として投稿、failure mode を主題化
3. **H1 FAILURE** (MAPE > 60%): “negative result” 論文として投稿、failure mode を発見化
4. **H7 reversal** ($\Delta P_B \leq \Delta P_A$ or ΔP_C): “thesis revision” 論文として投稿、original framing を修正
5. **B3 < B2** (typology overfitting): “critical finding” 論文として投稿、本研究の typology approach の限界を明示

7.2 Target journal path

> ***Note**: Pre-registration が lock するのは **analysis plan**（hypotheses、models、thresholds、sensitivity sweeps、ethics commitments）であり、**投稿先 journal は自由に変更可能**。Section 6.5 deviation policy にも該当しない。本 Section は **submission roadmap** として記録するのみ。*

Primary target

Royal Society Open Science (Registered Report track)

- 選定理由：(1) RR track で IPA 取得後は publication 保証 → D-NEW8 negative-result commitment が構造的に担保、(2) 予算内（£1,800 ≈ ¥35 万）、(3) Open Access 標準で D-NEW9 reproducibility と整合、(4) Stage 1 review ~6–8 週、(5) IF 制約なし（user 明示）
- Submission unit: **Stage 1 (Introduction + Methods + 本 pre-reg)** を提出 → IPA → Stage 2 (Results + Discussion) を後に提出

Fallback path (IPA refused または scope mismatch 時)

Order	Journal	Track	APC (¥)	採択確実性	備考
1	**Royal Society Open Science**	RR	¥35 万	RR IPA 後 guarantee	Primary
2	**Frontiers in Psychology** (Personality and Social Psychology specialty)	RR (一部 specialty)	¥44 万	RR specialty で同等	Fallback 1
3	**Scientific Reports** (Nature)	Standard with pre-reg DOI	¥34 万	~50%	Fallback 2、broad scope
4	**PLOS ONE**	Standard with pre-reg DOI	¥29 万	~50%、negative result accepting	Fallback 3
5	**PAID** (Personality and Individual Differences)	Standard	OA: ¥52 万 / 購読型: ¥0	~40%	Fallback 4、personality fit
6	**JBE** (Journal of Business Ethics)	Standard	OA: ¥58 万 / 購読型: ¥0	~10–20%	Fallback 5、ethics scope 厳
7	**RIO Journal**	Pre-Registration article type or Research Article	¥7–13 万	高	Fallback 6、低 APC 緊急時
8	**PCI Registered Reports + PCI-friendly OA journal**	RR (decentralized)	¥0	recommender 探し時間要	Fallback 7、cost-zero option

本 pre-reg は journal-agnostic

- 本 pre-reg の commitments は journal 選択に依存せず適用
- Journal 変更は Section 6.5 Level 0 (no deviation) として扱い、deviation log には記録不要
- 各 fallback は OSF pre-reg DOI を citable 引用として活用

7.3 Publication 拒否時の対応

- 全 fallback 失敗時：Pre-print (OSF / SocArXiv / PsyArXiv) で公開
- Failure mode の technical report を OSF supplementary として permanent 公開
- 本 pre-reg を最低 10 年間 OSF で active 維持

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8. Reproducibility Infrastructure (D-NEW9)

Anchor: **Munafò et al. 2017 Nature Human Behaviour** “A manifesto for reproducible science” 5 themes (Methods / Reporting / Reproducibility / Evaluation / Incentives) + TOP guidelines.

8.0 Munafò et al. 2017 5 themes への mapping

Theme	本研究での実装	Section
1. Methods（cognitive bias 防止 / methodology training / independent statistical oversight / team science）	(a) Blinding-equivalent: 14-cell cross-tabulation 生成前に本 pre-reg を OSF 登録（Section 2.2）、(b) Independent statistical oversight: 数理生物学者（mode B、anonymous）が Section 5 を Stage 2 着手前に review（Section 1.2 + Section 8.1）、(c) Methodology training / team science は sole-authored secondary analysis のため N/A	1.2, 2.2, 8.1
2. Reporting / Dissemination（pre-registration / reporting checklists CONSORT/ARRIVE/PRISMA/STROBE/STARD / COI 開示）	(a) OSF Standard Pre-Registration＝本ドキュメント、(b) Reporting は STROBE (observational study) と STARD（適用時）を Methods に統合（Section 8.1）、(c) COI 開示（Section 12.1）	8.1, 8.3, 12.1
3. Reproducibility（open data / materials / software / TOP guidelines / Registered Reports）	(a) Open code (MIT) を GitHub + OSF mirror、(b) Open aggregated data、再同定 risk のある raw のみ restricted access（Section 9.5）、(c) Random seed (20260429)、environment pinning (uv lock or Docker)、`make reproduce` 30 分目標、(d) **TOP guidelines**: 本研究は **Tier 3 / "Verification"** を Data Citation, Data Transparency, Analytic Methods Transparency, Research Materials Transparency, Design and Analysis Transparency, Preregistration of Studies, Preregistration of Analysis Plans, Replication の全項目で目指す、(e) Registered Reports 投稿（Section 7.2）	7.2, 8.1, 8.2, 9.5
4. Evaluation（preprints / post-publication review / results-blind review）	(a) **Results-blind review** は Registered Report track で達成（Section 7.2 RSOS RR primary）、(b) 全 venue refuse 時は preprint commitment（Section 7.3、OSF / SocArXiv / PsyArXiv）、(c) **Post-publication review**: 本 pre-reg と supplementary を OSF で 10 年以上 active 維持（Section 9.6）、misuse 報告に対し PubPeer / commentary 対応可能	7.2, 7.3, 9.6
5. Incentives（badges / Registered Reports / TOP guidelines / replication funding / hiring open practices）	(a) **OSF Open Practice Badges** — 本研究は Open Data, Open Materials, Preregistered の 3 badge すべて適格（aggregated data open、code open、analysis preregistered）、(b) Registered Report 投稿（Section 7.2）、(c) TOP-aligned reporting（本 Section 8.0 row 3）、(d) Replication funding / hiring は本 pre-reg では N/A だが Munafò 2017 が指摘する systemic lever として acknowledge	7.2, 8.0 row 3, 8.1

8.1 Pre-registered infrastructure commitments

Item	Implementation
Random seed	NumPy `default_rng(seed=20260429)`, Python `random.seed(20260429)`, Stan `seed=20260429`. Seed value 固定、本 pre-reg に明記。Bootstrap resample state は HDF5 で永続保存。
環境 pinning	(a) `uv` lock file (Python 3.11+) **OR** (b) Dockerfile (Python:3.11-slim base) のいずれかで完全 pinning
Make reproduce	`make reproduce` で全 figure / table / supplementary を 30 分以内に再生成（D-NEW9 README 必須項目）
Open data	Aggregated cell-level statistics (14-cell, 28-cell EB) は OSF / GitHub で完全公開。Cell-level raw data は restricted access (Section 9.5 ethics)
Open code	GitHub + OSF mirror。MIT license。
Reporting checklist	OSF Pre-Registration template + STROBE (observational study) + STARD (where applicable) を Methods に統合
Independent statistical oversight	Co-author / external methodologist による review を **Stage 1 完了後・Stage 2 着手前** に実施（Munafò 2017 Box 1 CHDI Foundation 例）

8.2 Pre-registered repository structure

```
simulation/
├── docs/
│   ├── pre_registration/
│   │   ├── D12_pre_registration_OSF.md (本ドキュメント、内部 master)
│   │   ├── D12_pre_registration_OSF.en.md (英訳版、OSF 提出用)
│   │   └── osf_registration_metadata.json (登録日時、DOI、version log)
│   ├── notes/
│   ├── literature_audit/
│   └── power_analysis/
├── code/
│   ├── stage0_type_assignment.py
│   ├── stage0_cell_propensity.py
│   ├── stage1_population_aggregation.py
│   ├── stage2_validation.py
│   ├── stage3_sensitivity.py
│   ├── stage4_baselines.py
│   ├── stage5_cmv_diagnostic.py
│   ├── stage6_target_trial.py
│   ├── stage7_counterfactual.py
│   └── stage8_transportability.py
├── tests/
│   └── test_*.py (each stage)
├── output/
│   ├── tables/
│   ├── figures/
│   └── supplementary/
├── Dockerfile
├── pyproject.toml + uv.lock
├── Makefile (with `reproduce` target)
└── README.md (with "How to reproduce in 30 minutes" section)
```

8.3 Pre-registered reporting items

論文 Methods + supplementary に必須記載:

- [ ] Random seed 値 (20260429)
- [ ] Software version: Python, NumPy, SciPy, scikit-learn, statsmodels, Stan/brms (適用時), PyMC (適用時)
- [ ] Hardware: CPU model, RAM, OS (再現性に影響する範囲)
- [ ] Bootstrap iterations: 2,000 (per cell)
- [ ] BCa CI parameters: jackknife-based bias z_0 + acceleration a の数値
- [ ] EB shrinkage hyper-parameters: $\hat{\alpha}$, β の point estimate と sensitivity sweep の各 scale
- [ ] MoM diagnostic: $\hat{\sigma}^2$ が $\hat{\mu}(1-\hat{\mu})$ に対してどの fraction かを report
- [ ] CMV diagnostic: Harman's first factor variance %, marker variable correction adjustments

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9. Ethical Commitments (D-NEW10、研究計画 v6 Part 7 全項目)

9.1 Pre-registered triple-locking 構造

Anchor: 研究計画 v6 Part 7（Sensitive topic としての harassment 研究の特性に対する 8 セクション対応）

Lock 位置	Content
(a) Methods 内	“Individual application disavowal” statement (Section 9.2 全文) を Methods 末尾に明記
(b) Discussion 内	“How this should NOT be used” 専用段落、Cluster X 個人を加害者と決めつけない言語選択遵守
(c) Pre-registration 内 (本 Section 9)	Part 7 全項目を ethical commitment として明示、本 pre-reg を OSF で公開

9.2 Individual application disavowal (Anti-screening Statement)

論文 Methods / Discussion / 著作の各所に以下相当の disclaimer を配置:

> ***Statement on individual application****: This study estimates* ***population-level statistical patterns****, not individual risk profiles. The cell-conditional probabilities reported here have* ***no validity for individual prediction****, screening, hiring, promotion, or any other personnel decision. Application of these findings to individual personnel decisions would constitute (a) statistical misuse, given the pairwise MDE of $d \geq 0.92$ and absent individual-level predictive validation, and (b) ethically unacceptable discrimination based on personality profiles. The authors explicitly disavow such application.*

加えて以下 4 点を併記:

- Cluster 0 でも加害確率は < 100%
- Cluster 1–6 でも加害は発生する
- Personality は変化可能（Roberts 2017）→ Cluster は固定属性ではない
- 採用・昇進判断への利用は研究倫理違反

9.3 Authorial framing guidelines (pre-registered)

避ける表現	推奨表現
“perpetrator personality”	“personality profile associated with elevated risk in our model”
“high-risk individuals”	“cell with elevated baseline rate (population-level pattern)”
“Cluster X should be screened”	“Cluster X represents a target group for **opt-in voluntary intervention**”
“predict perpetrators”	“predict population-level prevalence”
“effective intervention reduces harassment”	“anchor literature suggests intervention can reduce harassment under specified transportability assumptions”

9.4 Voluntary, opt-in 政策提言原則 (Counterfactual B 主軸の必須条件)

- **Voluntary participation**: 強制的研修・カウンセリングではなく opt-in
- **No employment consequence**: 参加の有無が雇用・評価に影響しない
- **Anonymity / Confidentiality**: 参加者の personality profile が組織内同定不能
- **Resource provision, not coercion**: 社会の役割は支援提供、強制ではない

9.5 Data availability ethics (D17)

Data tier	Public availability
Aggregated statistics (14-cell propensity, 28-cell EB, national prevalence, ΔP_x)	完全公開 (OSF + GitHub)
Cell-level raw data (cell ごとの個人 records)	Restricted access (要研究目的審査、IRB 確認、再同定 risk 評価) — 28-cell の N<10 cells で再同定 risk あり
N=354 / N=13,668 個人データ	既存論文の公開方針に準拠（個人特定情報削除済 anonymized data のみ要望ベース提供）

9.6 Long-term ethical monitoring (10-year commitment)

- **誤用 case の monitoring**: discrimination / screening 転用報告に対し correction / commentary 発表
- **二次解析 collaboration の選別**: ethical principle 適合審査
- **Author contact 維持**: corresponding author 連絡先 10 年以上維持
- **OSF maintenance**: 本 pre-reg と supplementary を 10 年以上 active 維持

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10. Limitations Pre-Acknowledged (研究計画 Part 4.2 11 limitations)

本 pre-reg で予め以下の limitation を acknowledge し、Discussion での post-hoc rationalization を回避（Nosek 2018 Challenge 9）：

#	Limitation	対応文献
L1	Self-report → Self-report 循環性	Berry et al. 2012, Anderson & Bushman 2002, Vazire 2010 SOKA (direction = conservative)
L2	N=354 代表性 (cloudsourcing self-selection)	Tsuno et al. 2015 N=1,546 random sample triangulation
L3	役職推定の誤差	D13 で continuous covariate 化、D1 で 3 モデル比較
L4	Cell-level 統計検出力 (pairwise MDE d ≥ 0.92)	D13 power analysis、aggregate inference 重視
L5	被害者倍率 V / 離職率 f / メンタル疾患率 f2 不確実性	Stage 3 sensitivity sweep
L6	Cross-sectional 因果不可能 + reverse causation	Roberts & DelVecchio 2000 plateau r=.74、Specht 2011 mid-life stability
L7	GAM situational 欠落	“Personality slice” framing、Bowling & Beehr 2006 acknowledged
L8	Common Method Bias	Podsakoff 2003 standard、Stage 5 diagnostic
L9	Phase 2 transportability (西欧 anchor → 日本)	Section 5.8 sensitivity sweep、Sapouna 2010 / Nielsen 2017
L10	Latent vs expressed prevalence 分離不可	Section 1.4 framing で「境界連続的」明示
L11	**Construct validity** (harassment scale が “latent propensity” を測っているか未検証)	Vazire 2010 SOKA、convergent / discriminant validity check (supplementary)

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11. Researcher Reflexivity (Positionality)

11.1 Pre-registered positionality statement

論文 Methods / Acknowledgments に以下を明記:

> *“The first author has previously argued in non-peer-reviewed work for a systemic-causation framing of workplace harassment, emphasizing social systems’ responsibility over individual self-responsibility (Tokiwa, forthcoming). This normative stance may influence which limitations are emphasized and how findings are framed in the Discussion. To mitigate, (a) the empirical analysis section is restricted to L1 descriptive/predictive claims, (b) all causal language is constrained by the target trial emulation framework (Hernán & Robins, 2020) with explicit identifying assumptions, (c) anti-screening and anti-discrimination statements are included regardless of findings, (d) limitations are presented in a structured 11-item framework derived from a 60+ paper literature audit (not selectively), (e) negative result publication is committed in advance via Section 7 of this preregistration (D-NEW8), and (f) independent methodologist review is sought prior to Stage 2 validation (Section 8.1 D-NEW9).”*

11.2 Bias mitigation procedures (pre-registered)

- Pre-registered analysis plan (本ドキュメント) で post-hoc 修正可能性を制限 (Nosek 2018)
- Co-author / external review for Discussion interpretation (Munafò 2017 Box 1)
- Failure mode の事前 commitment (Section 7)
- Self-report direction-of-bias awareness: Vazire 2010 SOKA 上 conservative (lower bound) として framing

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12. Other (Conflicts, Funding, Data Sources)

12.1 Conflicts of Interest

- 著者は本研究結果から商業的利益を受けない
- 著者所属（SUNBLAZE Co., Ltd.）は本研究結果に対する商業的利害関係を有しない
- 本研究は既存 IRB 通過済データの secondary analysis であり、新規データ収集なし
- Pre-registration の OSF 登録は無料、author cost なし

12.2 Funding

- **Simulation phase（本研究）**: No external funding. 著者所属（SUNBLAZE Co., Ltd.）の通常業務時間内で実施。
- **Original data collection**:
  - N=354 (harassment data): Tokiwa harassment preprint で記載済の元 IRB / funding を継承
  - N=13,668 (clustering data): Tokiwa clustering paper (IEEE 掲載済) で記載済の元 IRB / funding を継承
- **新規 IRB**: 不要（本研究は simulation のみで人を対象とする新規研究行為を含まない、既存 anonymized data の secondary analysis）

12.3 Data Sources Summary

Source	Type	Access
`harassment/raw.csv` (N=354)	Primary	Authors' prior IRB-approved collection (Tokiwa harassment preprint)
`clustering/csv/clstr_kmeans_7c.csv`	Derived	Tokiwa clustering paper IEEE 掲載済 (centroid table)
MHLW H28 (FY2016) 職場のパワーハラスメント実態調査	Public	https://www.mhlw.go.jp/ (pre-law, 32.5%, ★ primary validation target H1)
MHLW R2 (FY2020) 職場のハラスメント実態調査 報告書	Public	https://www.mhlw.go.jp/ (transition, 31.4%)
MHLW R5 (FY2023, published March 2024) 令和5年度 厚生労働省委託事業 職場のハラスメントに関する実態調査報告書 (PwC コンサルティング合同会社, 385 pages)	Public	`simulation/prior_research/_text/令和５年度_厚生労働省委託事業_職場のハラスメントに関する実態調査報告書.pdf` (post-law, 19.3%, full report; industry-stratified data 16-26.8% used in B4 H2.industry validation)
MHLW R4 (FY2022) 雇用動向調査結果の概況	Public	`simulation/prior_research/_text/厚生労働省_令和４年雇用動向調査結果の概況.pdf` (FY2022 turnover by reason "interpersonal" 8.3-9.4%; f1 secondary anchor)
MHLW 労働安全衛生調査	Public	https://www.mhlw.go.jp/
MHLW 労働力調査	Public	https://www.stat.go.jp/data/roudou/ (industry × demographic crosstabs for B4 industry estimation, Section 5.5)
ILO 2022 Global survey	Public	https://www.ilo.org/
Tsuno et al. 2015 N=1,546	Published	Tsuno et al. 2015 PLOS ONE (national-rep 30-day prevalence 6.1%)
Tsuno & Tabuchi 2022	Published	Tsuno & Tabuchi 2022 (PR=3.20 for f2 anchor)
Pasona Research (2022) 職場のハラスメントについての定量調査 (株式会社パーソル総合研究所, シンクタンク本部, 152 pages)	Industry survey	`simulation/prior_research/_text/パーソル_職場のハラスメントについての定量調査.pdf` (★ N=28,135 large-N harassment-specific 5-year prevalence 19.7%, 業種別 16.9-22.9%; primary f1 anchor 10.3% [業種別 6.3-13.3%]; macro turnover estimate 86.5万人/年 with 暗数 66%)

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13. Citation Anchors (Tier 1+2+3+4 文献基盤)

本 pre-reg の各 commitment は以下文献に依拠。完全 list は `simulation/docs/literature_audit/deep_reading_notes.md` (40 paper deep reading) を参照。

13.1 Tier 4 metascience anchors (★ 本 pre-reg の核)

- **Nosek, Ebersole, DeHaven & Mellor (2018)**. The preregistration revolution. *PNAS, 115*(11), 2600–2606. https://doi.org/10.1073/pnas.1708274114 [→ Section 0.2, 3.1, 7]
- **Munafò et al. (2017)**. A manifesto for reproducible science. *Nature Human Behaviour, 1*, 0021. https://doi.org/10.1038/s41562-016-0021 [→ Section 8]
- **Vazire (2010)**. Who knows what about a person? The Self–Other Knowledge Asymmetry (SOKA) Model. *JPSP, 98*, 281–300. [→ Section 10 L1, L11]
- **Funder & Ozer (2019)**. Evaluating effect size in psychological research: Sense and nonsense. *AMPPS, 2*, 156–168. [→ Section 3.3]

13.2 Causal inference framework

- **Hernán & Robins (2020)**. Causal Inference: What If. Chapman & Hall/CRC. [→ Section 5.7.4]
- **Pearl (2009)**. Causality: Models, Reasoning, and Inference (2nd ed.). Cambridge. [→ Section 5.7.2]

13.3 Statistical methods anchors

- **Efron (1987)**. Better bootstrap confidence intervals. *J Am Stat Assoc*. [→ BCa]
- **DiCiccio & Efron (1996)**. Bootstrap confidence intervals. *Stat Sci*. [→ BCa]
- **Casella (1985)**. An introduction to empirical Bayes data analysis. *Am Stat*. [→ EB]
- **Clayton & Kaldor (1987)**. Empirical Bayes estimates of age-standardized relative risks. *Biometrics*. [→ EB epidemiology]
- **Efron (2014)**. Two modeling strategies for empirical Bayes estimation. *Stat Sci*. [→ EB modern]
- **Greenland (2000)**. Principles of multilevel modelling. *Int J Epidemiol*. [→ EB / multilevel]
- **Podsakoff, MacKenzie, Lee & Podsakoff (2003)**. Common method biases in behavioral research. *J Appl Psychol*. [→ CMV]
- **Lindell & Whitney (2001)**. Accounting for common method variance. *J Appl Psychol*. [→ marker variable]
- **Schofield et al. (2018)**. Health-related microsimulation. *Eur J Health Econ*. [→ MAPE microsim]

13.4 Phase 2 intervention anchors

- **Kruse, Chancellor, Ruberton & Lyubomirsky (2014)**. An upward spiral between gratitude and humility. *Soc Psychol Personal Sci, 5*(7), 805–814. [→ Counterfactual A]
- **Hudson (2023)**. Lighten the darkness: Personality interventions targeting agreeableness. *J Personality, 91*(4). [→ Counterfactual B 主軸]
- **Pruckner & Sausgruber (2013)**. Honesty on the streets. *J Eur Econ Assoc, 11*(3), 661–679. [→ Counterfactual C]
- **Bezrukova et al. (2016)**. Diversity training meta-analysis. [→ Counterfactual C triangulation]
- **Roehling & Huang (2018)**. Sexual harassment training meta-analysis. [→ Counterfactual C triangulation]
- **Dobbin & Kalev (2018)**. 985 studies meta. [→ Counterfactual C triangulation]
- **Roberts et al. (2017)**. Personality trait change through intervention systematic review. *Psychol Bull, 143*(2), 117–141. [→ 24-week duration anchor]

13.5 Personality + harassment anchors

- **Pletzer et al. (2019)**. HEXACO meta. [→ HH × CWB ρ ≈ -.20 to -.35]
- **Nielsen, Glasø & Einarsen (2017)**. FFM × harassment meta. [→ cultural moderator]
- **Bowling & Beehr (2006)**. Harassment from victim's perspective meta. [→ environmental ρ]
- **Roberts & DelVecchio (2000)**. Rank-order consistency meta. [→ stability r=.74]
- **Specht et al. (2011)**. SOEP N=14,718 stability. [→ mid-life stability]
- **Ashton & Lee (2007)**. HEXACO model. [→ trait taxonomy]
- **Wakabayashi (2014)**. Japanese HEXACO-60. [→ Japanese measurement]
- **Tou et al. (2017)**. Workplace Power Harassment Scale. [→ measurement]
- **Kobayashi & Tanaka (2010)**. Gender Harassment Scale. [→ measurement]
- **Shimotsukasa & Oshio (2017)**. SD3-J. [→ Dark Triad measurement]
- **Lee & Ashton (2005)**. HEXACO ↔ Dark Triad correlations. [→ trait inter-correlation]
- **Berry, Carpenter & Barratt (2012)**. CWB self-other meta. [→ self-report defense]

- **Anderson & Bushman (2002)**. GAM. [→ self-report → real behavior]

13.6 Personality upstream of SES

- **Heckman, Stixrud & Urzua (2006)**. Noncognitive skills predict outcomes. [→ Section 1.5]
- **Grijalva et al. (2015)**. Narcissism → leadership emergence meta. [→ Section 1.5]
- **Roberts et al. (2007)**. The power of personality. [→ Section 1.5]

13.7 Japanese context

- **MHLW (H28, FY2016)** 平成28年度 職場のパワーハラスメントに関する実態調査. [→ ★ H1 primary validation target, 32.5% pre-law power harassment past-3-year]
- **MHLW (R2, FY2020, published 2021)** 令和2年度 職場のハラスメントに関する実態調査 報告書. [→ secondary validation 31.4% transition]
- **MHLW (R5, FY2023, published March 2024)** 令和5年度 厚生労働省委託事業 職場のハラスメントに関する実態調査報告書 (PwC コンサルティング合同会社, 385 pages). [→ ★ secondary validation 19.3% post-law; ★ industry-stratified data 16-26.8% used in H2.industry secondary criterion (Section 5.5, 6.1); カスハラ 10.8% as new environmental category (Section 1.4 H3)]
- **MHLW (R4, FY2022, published August 2023)** 令和4年雇用動向調査結果の概況. [→ ★ f1 secondary anchor: turnover by reason "interpersonal" 男 8.3% / 女 9.4% as upper bound]
- **Pasona Research (2022)** 職場のハラスメントについての定量調査 (株式会社パーソル総合研究所, シンクタンク本部, N=28,135 全国就業者 20-69 歳, 152 pages). [→ ★ Pre-reg validation triangulation (Section 5.4): 5-year harassment prevalence 19.7%, industry-stratified 16.9-22.9%; ★ f1 PRIMARY empirical anchor: harassment-victim turnover rate 10.3% (industry range 6.3-13.3%) (Section 4.2.3, 6.4); ★ Stage 2 chain output sanity check (Section 5.4): macro turnover 86.5万/年 with 66% 暗数化]
- **Tsuno et al. (2010)** Japanese NAQ-R. [→ measurement]
- **Tsuno et al. (2015)** Socioeconomic determinants Japan national rep N=1,546 (PLOS ONE). [→ Section 1.5, 4.2; 30-day prevalence 6.1% as reference]
- **Tsuno & Tabuchi (2022)**. Bullying → SPD PR=3.20. [→ f2 anchor]

13.8 Self-citation hub

- **Tokiwa et al.** Clustering paper (IEEE 掲載済). [→ N=13,668 7-type centroid]
- **Tokiwa et al.** Harassment preprint. [→ N=354 HEXACO + Dark Triad regression]

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14. Version Log & Implementation Checklist

14.1 Version log

Version	Date	Changes
v1.0 draft	2026-04-29	Initial draft based on research plan v6/v7 (1,458 行) + D13 power analysis (209 行) + 40 paper deep reading. Pending OSF registration.
v1.1 draft	2026-04-29	Pre-OSF-lock revision integrating 3 new domestic surveys uploaded by author: (1) MHLW R5 (FY2023, 385 pages, full report) — used for industry-stratified validation in H2.industry secondary criterion (Section 5.5, 6.1); (2) MHLW R4 (FY2022) Employment Trend Survey — used for f1 secondary anchor; (3) Pasona Research (2022) N=28,135 quantitative survey — used for Pasona triangulation (Section 5.4), f1 PRIMARY empirical anchor (Section 4.2.3, 6.4 — value 10.3% with industry range 6.3-13.3%), Stage 2 chain output sanity check (predicted annual harassment-induced turnover should fall within 50-200% of Pasona estimate 86.5 万人/年). All MHLW citations standardized: era code (H28/R2/R5) + fiscal year + scope (past-3-year power harassment) + policy phase (pre-law/transition/post-law). Added カスハラ category emergence (10.8% in

		MHLW R5) as evidence of environmental moderation in latent vs expressed framing (Section 1.4 H3). H1 main MAPE threshold (≤30%) and sensitivity ranges UNCHANGED; only secondary criteria (H2.industry, Stage 2 chain sanity check) and citation precision improved. Pending OSF registration.
v1.1 LOCKED	2026-04-30	🔒 Registered on OSF. **DOI: 10.17605/OSF.IO/45QP9** (https://osf.io/45qp9). Associated project: https://osf.io/3hxz6. Subsequent modifications must follow Section 6.5 Level 3 deviation procedure (v2 registration with public diff against v1.1). Stage 0 code execution was unlocked at this point but deferred pending methodologist review.
v2.0 draft	2026-04-30	Path C upgrade following anonymous methodologist review (Matsuda mode B, mathematical biology background). Four substantive changes: (1) **4-tier judgment hierarchy** in Section 5.4 / 6.1 — Strict SUCCESS = point MAPE ≤ 30% AND 95% BCa CI upper bound ≤ 30%; Standard SUCCESS = point MAPE ≤ 30% with explicit CI ambiguity Tier qualifier required; PARTIAL / FAILURE unchanged. (2) **Bootstrap MAPE CI computation procedure** pre-specified in Section 5.4 (cell-stratified resample → re-classify with fixed centroids → re-estimate cell propensity → apply fixed population weights → compute MAPE^(b) → 95% BCa CI from empirical distribution; population weights and MHLW observed values fixed across iterations to isolate cell-level uncertainty). (3) **MoM rejection decision rule** in Section 5.2 — REJECT MoM and switch to Stan hierarchical Bayes if $\hat{\sigma}^2 / [\hat{\mu}(1-\hat{\mu})] < 0.05$ OR $\max(\hat{\alpha}, \hat{\beta}) > 100$; alternative diagnostic ($\hat{\alpha} + \hat{\beta}$ as fraction of median cell N=14) reported as cross-check. (4) **BCa numerical stability fallback chain** in Section 5.1 — BCa → BC (bias-corrected) → percentile bootstrap, triggered by
v2.0 LOCKED	2026-04-30	🔒 Registered on OSF as v2 with public diff (D12_pre_registration_v1.1_to_v2.0_diff.pdf) attached as supplementary. **v2.0 OSF DOI: 10.17605/OSF.IO/3Y54U** (https://osf.io/3y54u). Associated project: https://osf.io/3hxz6. v1.1 (DOI 10.17605/OSF.IO/45QP9) remains valid as historical record but is superseded by v2.0 for primary inference. Stage 0 code execution unlocked under v2.0 specifications.

14.2 Pre-registration submission checklist

14.2.1 OSF pre-reg 登録（独立した track）

- [N/A] **Internal review**: Co-author confirmation — **sole-authored study につき不要**
- ☐ **Self-review against Nosek 2018 9 challenges**: Section 0.2 全埋
- ☐ **Self-review against Munafò 2017 5 themes**: Section 8 全埋
- ☐ **English translation**: `D12_pre_registration_OSF.en.md` 作成
- ☒ **OSF account**: 既存 (metaanalysis/ で使用) → **新規 project 作成のみ pending**
- ☒ **OSF Standard Pre-Registration template**: Sections 1–6 を OSF Web フォームに転記済（Part B paste sheet 使用）
- ☒ **PDF supplementary**: JP/EN markdown + JP/EN PDF を OSF に attach 済

- ☒**DOI 取得**: 10.17605/OSF.IO/45QP9 取得済、本ドキュメント Header に記録済
- ☒**GitHub mirror**: `simulation/docs/pre_registration/` を public commit (commit `c5c591e` 以降で push 済)
- ☒**Funding & affiliation**: Section 12.2 fill-in 完了（SUNBLAZE Co., Ltd. / no external funding for simulation phase）
- ☐**Anti-screening triple-lock**: Section 9.1 三箇所すべて準備済
- ☐**Independent methodologist contact**: Section 8.1 review 依頼確定（mode B: anonymous, math bio）

14.2.2 Registered Report 投稿（pre-reg lock 後の別 track）

☐ RSOS Registered Report **Author Guidelines** 確認（最新版、scope / format requirement）
☐ **Stage 1 manuscript** 準備（Introduction + Methods + 本 pre-reg を統合した英文原稿、Stage 1 用 word limit 内）
  - Introduction: 研究計画 v6/v7 Part 10.3 の 5 段落構成を base に英訳
  - Methods: 本 pre-reg Section 4–5 を base に英訳、computational reproducibility (D-NEW9) を含む

☐ **Cover letter** 作成（既存 OSF pre-reg DOI 参照、IPA 取得を依頼）
☐ **Submission portal**（RSOS / Editorial Manager 等）への投稿
☐ **IPA review 対応**（typically 6–8 週、reviewer comments への revision）
☐ **IPA acquired** → Stage 0 コード実装着手（Section 14.3）
☐ **Stage 2 manuscript**（Results + Discussion）作成、IPA を base に Stage 2 投稿
☐ **Final acceptance** → publication

14.2.3 Fallback path（IPA refused 時、Section 7.2 fallback table 参照）

☐ Frontiers in Psychology RR specialty に再投稿、または
☐ Standard track に切替（Stage 0–8 完了後に Sci Reports / PLOS ONE 等に投稿）
☐ OSF pre-reg DOI を必ず引用、deviation log を transparent に記載

14.3 Stage 0 着手前の最終確認

以下がすべて完了した時点で本 pre-reg を **lock**、Stage 0 コード実装を開始:

- ☒OSF DOI 取得済（v1.1: 10.17605/OSF.IO/45QP9; **v2.0 (current primary): 10.17605/OSF.IO/3Y54U**）
- ☒本ドキュメントの Header に DOI 追記済 (v2.0 LOCKED status 反映)
- ☒**Methods Clarifications Log v1.0** lock 済（2026-05-21、`D12_v2.0_methods_clarifications.md`、Section 6.5 Level 1 deviation log として OSF v2.0 project に supplementary upload 予定。Major×4 + Minor×7 + Nitpicks×5 を反映、5th tier proposal は SKIP）
- [N/A] Co-author sign-off — sole-authored につき不要
- ☐Independent methodologist による Section 5 (Analysis Plan) review 完了
- ☐Repository structure (Section 8.2) 初期化
- ☐ `make reproduce` skeleton 作成済
- ☐Random seed (20260429) を全 stage で hard-code 確認

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15. End-of-Document Statement

本ドキュメントは Pre-registration の **internal master draft** である。OSF への正式登録 (DOI 取得) を以て本 pre-reg は **lock** され、Stage 0 コード実装が解禁される。本ドキュメントの修正は Section 6.5 の Level 3 (analysis plan revision) 手続に従う。

Lock 状態: ⏳ DRAFT (pending internal review and OSF submission) **Next step**: Section 14.2 checklist の internal review item から順次実施

End of D12 OSF Pre-Registration v1.0 draft.